ic_autolab round 36 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-10T03:59:40Z at git `b46d1ee5fbc8507c95a6256b0982aad3180f55f7`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: none yet. Record one with `python3 tools/pcc/pcc.py claim new` before citing this round; until then its outcome is an unverified assertion [pcc:unverified].

Outcome

k class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
20 **null**	tw2l-x10-j16-n2	0.998 [0.931, 1.069]	0.977 [0.911, 1.048]	0.883 [0.819, 0.953]	sel < inc	2.446 / 2.627 / 4.257	2.452 / 2.689 / 4.821	1.492 / 1.598 / 2.551	1.6x / 1.6x / 1.7x
24 **improvement**	tw2l-x20-j32-n2	0.924 [0.857, 0.996]	0.924 [0.857, 0.998]	0.928 [0.856, 1.006]	sel < inc	2.016 / 2.189 / 3.743	2.182 / 2.368 / 4.034	1.125 / 1.226 / 2.133	1.8x / 1.8x / 1.8x
28 **null**	tw2l-x30-j16-n2	0.958 [0.869, 1.055]	0.959 [0.871, 1.058]	0.967 [0.875, 1.067]	sel < inc	1.768 / 1.938 / 3.473	1.846 / 2.020 / 3.593	1.016 / 1.111 / 1.968	1.7x / 1.7x / 1.8x
32 **null**	tw2l-x20-j8-n2	0.942 [0.843, 1.054]	0.925 [0.825, 1.037]	0.852 [0.751, 0.967]	sel < inc	1.772 / 1.948 / 3.529	1.880 / 2.106 / 4.139	1.005 / 1.104 / 1.993	1.8x / 1.8x / 1.8x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

- k = 20: compare ops.status = ok, admissible = true; sel / inc = 0.9977 [0.931, 1.069] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1
- k = 24: compare ops.status = ok, admissible = true; sel / inc = 0.9238 [0.857, 0.996] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1
- k = 28: compare ops.status = ok, admissible = true; sel / inc = 0.9579 [0.869, 1.055] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1
- k = 32: compare ops.status = ok, admissible = true; sel / inc = 0.9425 [0.843, 1.054] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1

Train (development screen, in sample)

k = 20: session `ECBS1h1c562b28d436` (run exit 0, verify exit 0, receipt sha256 `cf51440c88d1ca13266578c7cdab74e3eb5459429ba9cc41c41393fd08b8780d`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	- -	`rho.negation`	16	16	1.694 / 1.819 / 2.948	matched reference
`inc`	wsum2g-b15	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=15,free=1`	16	16	2.574 / 2.823 / 5.063	incumbent
`tw2l-x10-j8-n2`	tw2l-x10-j8	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wwalk:x=1.0,jumps=8,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run`	16	16	2.403 / 2.592 / 4.289	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
`tw2l-x10-j16-n2`	tw2l-x10-j16	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wwalk:x=1.0,jumps=16,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run`	16	16	2.284 / 2.448 / 3.925	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
`tw2l-x15-j16-n2`	tw2l-x15-j16	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wwalk:x=1.5,jumps=16,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run`	16	16	2.352 / 2.520 / 4.038	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
`tw2l-x20-j8-n2`	tw2l-x20-j8	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wwalk:x=2.0,jumps=8,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run`	16	16	2.363 / 2.545	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x =

arm	family	N factor	base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
							/ 4.179	2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
tw2l-x20-j16-n2	tw2l-x20-j16	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=2.0,jumps=16,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	2.487 / 2.667 / 4.287	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
tw2l-x20-j32-n2	tw2l-x20-j32	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=2.0,jumps=32,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	2.642 / 2.810 / 4.322	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
tw2l-x30-j16-n2	tw2l-x30-j16	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=3.0,jumps=16,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	2.429 / 2.603 / 4.168	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)

k = 24: session `ECBS1h11d09a8c4546` (run exit 0, verify exit 0, receipt sha256 `4cd17605ffe0b1d7a9527ce7b30fca23b99887a416792281562e3482616f884a`)

arm	family	N factor	base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
rho-neg	rho	-	-	rho.negation	16	16	1.409 / 1.538 / 2.702	matched reference
inc	wwalkGg-x09-sr	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.9,g=1,grow=2,seed=run	16	16	2.218 / 2.408 / 4.117	incumbent
tw2l-x10-j8-n2	tw2l-x10-j8	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0,jumps=8,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	1.817 / 1.982 / 3.469	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
tw2l-x10-j16-n2	tw2l-x10-j16	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0,jumps=16,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	2.185 / 2.383 / 4.169	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
tw2l-x15-j16-n2	tw2l-x15-j16	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.5,jumps=16,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	1.972 / 2.149 / 3.740	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
tw2l-x20-j8-n2	tw2l-x20-j8	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=2.0,jumps=8,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	1.992 / 2.174	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x =

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
						/ 3.814	2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
tw2l-x20-j16-n2	tw2l-x20-j16	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=2.0,jumps=16,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	2.194 / 2.393 / 4.178	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
tw2l-x20-j32-n2	tw2l-x20-j32	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=2.0,jumps=32,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	1.773 / 1.921 / 3.258	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
tw2l-x30-j16-n2	tw2l-x30-j16	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=3.0,jumps=16,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	1.956 / 2.131 / 3.703	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)

k = 28: session `ECBS1hfddd88f5cc82` (run exit 0, verify exit 0, receipt sha256 `c0c07530c90c8469907361226d229a99ba185f0f0b352a153a80d0cd1101d248`)

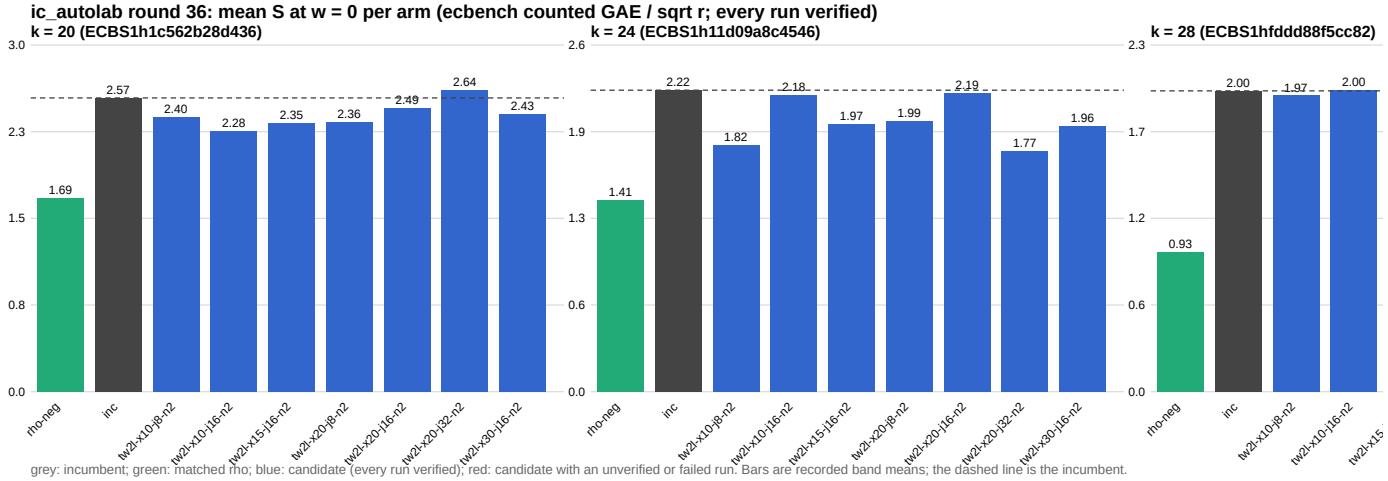
arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
rho-neg	rho	- -	rho.negation	16	16	0.929 / 1.016 / 1.794	matched reference
inc	wwalkGg-x09-sr	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.9,g=1,grow=2,seed=run	16	16	1.997 / 2.187 / 3.896	incumbent
tw2l-x10-j8-n2	tw2l-x10-j8	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0,jumps=8,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	1.965 / 2.157 / 3.881	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
tw2l-x10-j16-n2	tw2l-x10-j16	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0,jumps=16,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	2.001 / 2.195 / 3.942	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
tw2l-x15-j16-n2	tw2l-x15-j16	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.5,jumps=16,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	1.726 / 1.893 / 3.391	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
tw2l-x20-j8-n2	tw2l-x20-j8	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=2.0,jumps=8,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	1.688 / 1.852	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x =

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
						/ 3.325	2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
tw2l-x20-j16-n2	tw2l-x20-j16	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=2.0,jumps=16,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	1.678 / 1.840 / 3.293	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
tw2l-x20-j32-n2	tw2l-x20-j32	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=2.0,jumps=32,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	1.928 / 2.113 / 3.772	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
tw2l-x30-j16-n2	tw2l-x30-j16	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=3.0,jumps=16,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	1.614 / 1.770 / 3.166	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)

k = 32: session `ECBS1h406478868790` (run exit 0, verify exit 0, receipt sha256 `4321ebe0b1695edd4d09b9cddd8f644ccb1e566b0a929a001e7a2837e5dac9fa`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
rho-neg	rho	- -	rho.negation	16	16	1.138 / 1.250 / 2.259	matched reference
inc	wwalkG-x06-sr	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.6,g=1,grow=2,seed=run	16	16	1.807 / 2.020 / 3.942	incumbent
tw2l-x10-j8-n2	tw2l-x10-j8	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0,jumps=8,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	1.875 / 2.061 / 3.734	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
tw2l-x10-j16-n2	tw2l-x10-j16	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0,jumps=16,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	1.643 / 1.806 / 3.269	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
tw2l-x15-j16-n2	tw2l-x15-j16	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.5,jumps=16,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	1.893 / 2.080 / 3.768	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
tw2l-x20-j8-n2	tw2l-x20-j8	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=2.0,jumps=8,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	1.621 / 1.781	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x =

arm	family	N	factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
							/ 3.226	2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
tw2l-x20-j16-n2	tw2l-x20-j16	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=2.0,jumps=16,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	1.865 / 2.049 / 3.711	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
tw2l-x20-j32-n2	tw2l-x20-j32	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=2.0,jumps=32,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	1.712 / 1.880 / 3.399	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)
tw2l-x30-j16-n2	tw2l-x30-j16	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=3.0,jumps=16,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run	16	16	1.962 / 2.156 / 3.906	x/jumps sweep of tw2cj-lazy after the SNC panel (PCC-20261110-978c29: x = 2.0, 16 jumps gave 0.936/0.914/0.909/0.922 of the incumbent at k = 20/24/28/32)



train screen

Selection

- k = 20: challenger `tw2l-x10-j16-n2` (Pareto alternative at w = 1: `tw2l-x10-j16-n2`)
- k = 24: challenger `tw2l-x20-j32-n2` (Pareto alternative at w = 1: `tw2l-x20-j32-n2`)
- k = 28: challenger `tw2l-x30-j16-n2` (Pareto alternative at w = 1: `tw2l-x30-j16-n2`)
- k = 32: challenger `tw2l-x20-j8-n2` (Pareto alternative at w = 1: `tw2l-x20-j8-n2`)

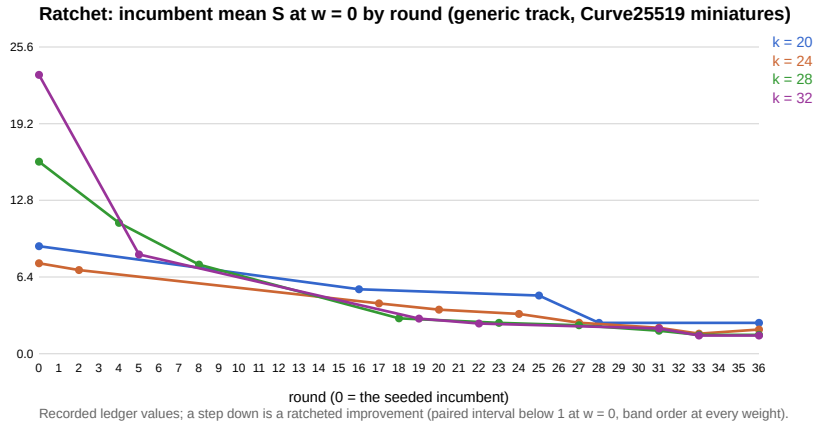
Holdout sessions

k	session	run exit	verify exit	receipt sha256	levels
20	ECBS1h4d22b97cc821	0	0	9d2d23404316557c1138c5f729b0532e0a45159f974efa8fdc54d872b1f3ca36	{“verified”:480}
24	ECBS1hbefb5a60befd	0	0	9c7f9c918ae01d295cc7dd4dc6bb46386c864ce3d4699a1cd70a14010e1e7a68	{“verified”:480}
28	ECBS1h464c8731ec2c	0	0	73a0b4830bb6724fafa6f89ee16617ff2059a2dee51effa5b278e29ced9466bb	{“verified”:480}
32	ECBS1h4f6f7a8be11f	0	0	f3000600d494cff45e88c84ba83bae8d56d4e2e54edf1756d4e947d4623e14cb	{“verified”:480}

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	wsum2g-b15-n2 (prime-projected-abscissa:size=2,scan_cap=32 + wsum:coef=15,free=1)	28	2.575 / 2.691 / 3.741	1.8x / 1.7x / 1.5x

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
24	`tw2l-x20-j32-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wwalk:x=2.0,jumps=32,g=1,grow=2,tw=2,cj=1,lazy=1,seed=run`)	36	2.016 / 2.189 / 3.743	1.8x / 1.8x / 1.8x
28	`wwalkGg-x09-sr-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wwalk:x=0.9,g=1,grow=2,seed=run`)	33	1.883 / 1.926 / 2.309	1.8x / 1.7x / 1.1x
32	`wwalkG-x06-sr-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wwalk:x=0.6,g=1,grow=2,seed=run`)	33	1.842 / 1.932 / 2.740	1.8x / 1.7x / 1.4x



ratchet

Evidence and deviations

- ecbench binary sha256: `1163f6b35d27ec8853455f6d88dfb9fb1d824745068df4dfd065bf6ada23b859`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.